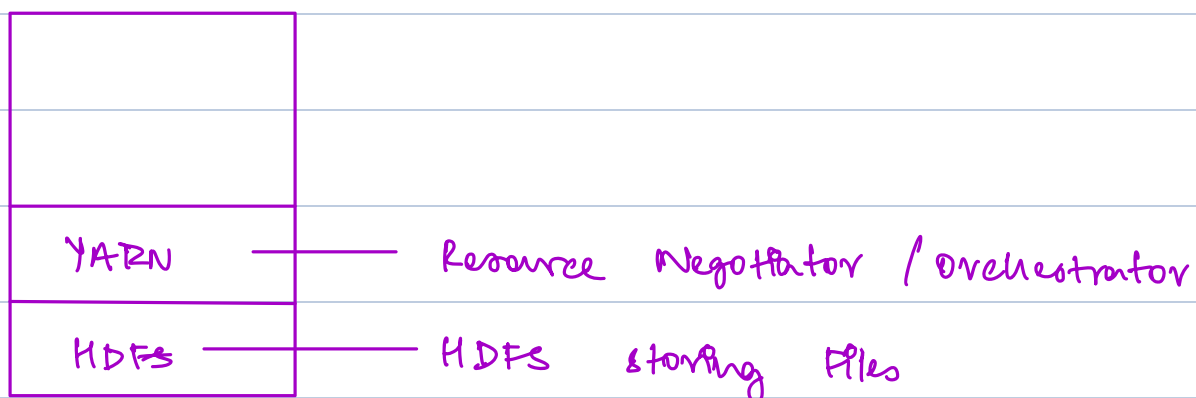


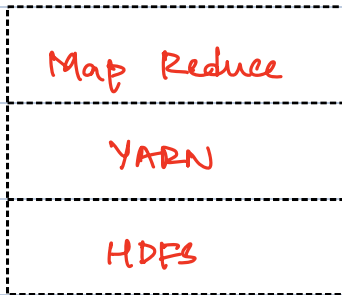


Agenda

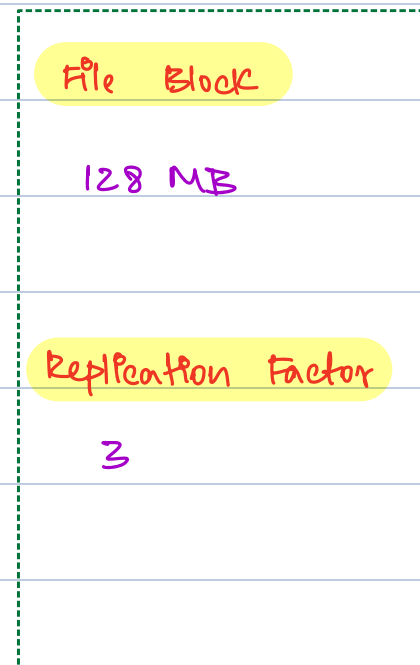
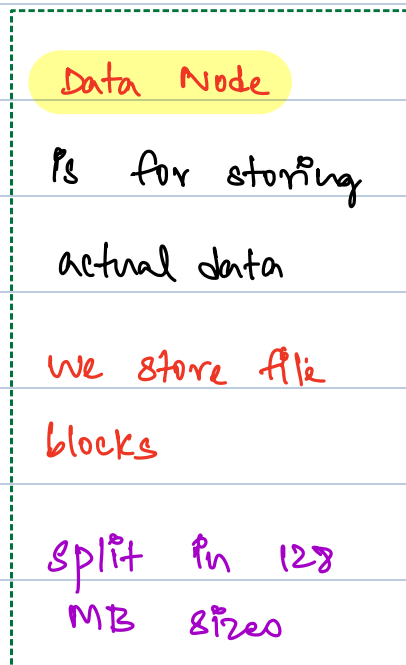
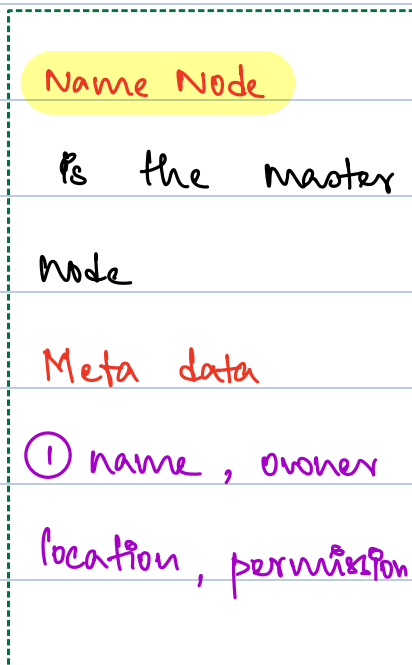
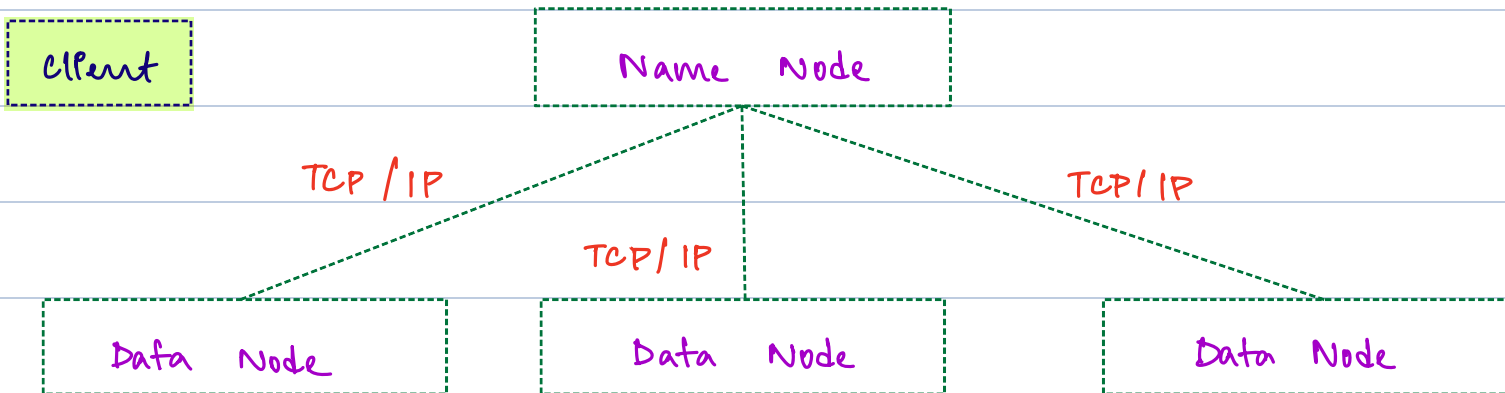
- ① Revision of HDFS
- ② Revision of YARN
- ③ Use case
- ④ Map Reduce



Revision of Hadoop Framework



Hadoop Distributed File System



Key Points to Remember

Yet another resource Negotiator



client



Resource Manager

Project Manager

TL

SE, SDET

Infosys

client



Job

Resource Manager

Scheduler

App Manager

Master Node

Container

App Master

Node Manager

Container

App Master

Node Manager

Yarn → Master Slave Architecture

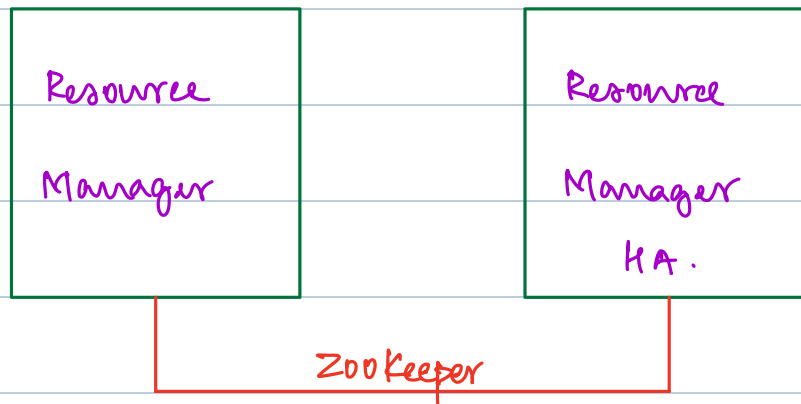
Master - Resource Manager

Slave - Node Manager

Communication Protocol - TCP/IP

Heartbeat - sent from Node manager to Resource Manager

Resource Manager HA → modern Architecture



Resource Manager



All Applications

Logged in as: dr who

Cluster

About Nodes

Node Labels

Applications

NEW

NEW SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

Cluster Metrics

Apps Submitted	0	Apps Pending	1	Apps Running	0	Apps Completed	2	Containers Running	3 GB	Memory Used	3 GB	Memory Total	0 B	Memory Reserved	2	VCores Used	2	VCores Total	0	VCores Reserved	
----------------	---	--------------	---	--------------	---	----------------	---	--------------------	------	-------------	------	--------------	-----	-----------------	---	-------------	---	--------------	---	-----------------	--

Cluster Nodes Metrics

Active Nodes	0	Decommissioning Nodes		Decommissioned Nodes		Lost Nodes	0	Unhealthy Nodes	0	Rebooted Nodes		Shutdown Nodes	0
--------------	---	-----------------------	--	----------------------	--	------------	---	-----------------	---	----------------	--	----------------	---

Scheduler Metrics

Scheduler Type	Capacity Scheduler	Scheduling Resource Type	[MEMORY]	Minimum Allocation	<memory:256, vCores:1>	Maximum Allocation	<memory:3072, vCores:1>	Maximum Cluster Application Priority	0
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Show 20 entries

Search:

ID	User	Name	Application Type	Queue	Application Priority	StartTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU VCores	Allocated Memory MB	% of Queue	% of Cluster	Progress	Tracking UI	Blacklisted Nodes
application_1508428570624_0001	username	Spark shell	SPARK	default	0	Thu Oct 19 08:58:23 -0700 2017	N/A	RUNNING	UNDEFINED	2	2	3072	100.0	100.0		ApplicationMaster	0

Showing 1 to 1 of 1 entries

First Previous 1 Next Last

Roles and Responsibility

① It keeps track of all meta data for jobs submitted by the user

It does not run the Business logic
It just orchestrates it

Scheduler

Just allocate resources based on scheduling algorithm

① FIFO

Takes first job
from queue, gives
all resources for
completion

② capacity (default)

Individual SLA's
will receive
dedicated resources

③ Fair → Lmb

I will have
predefined slot
based on which
I share resources

Application Manager

Manages all metadata about slaves and jobs submitted

→ Node Manager (Job, Resources, Jobs Active
in Node Manager)

→ This creates your first instance of
Application master for a given job

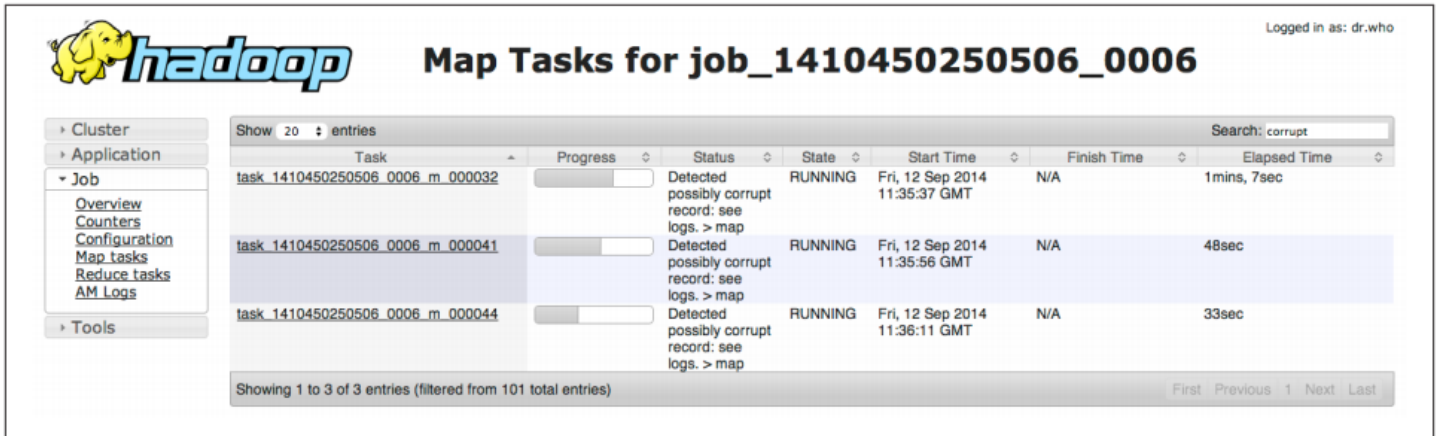
Container → Node Manager slave

slots

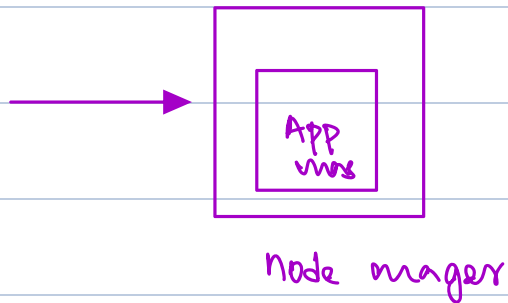
CPU + Memory blocks

Application Master

orchestrator of job

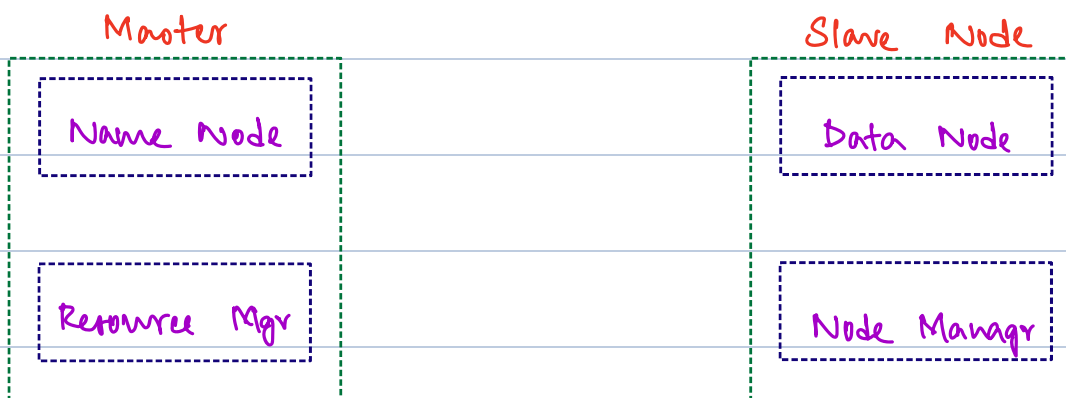


Task	Progress	Status	State	Start Time	Finish Time	Elapsed Time
task_1410450250506_0006_m_000032		Detected possibly corrupt record: see logs. > map	RUNNING	Fri, 12 Sep 2014 11:35:37 GMT	N/A	1mins, 7sec
task_1410450250506_0006_m_000041		Detected possibly corrupt record: see logs. > map	RUNNING	Fri, 12 Sep 2014 11:35:56 GMT	N/A	48sec
task_1410450250506_0006_m_000044		Detected possibly corrupt record: see logs. > map	RUNNING	Fri, 12 Sep 2014 11:36:11 GMT	N/A	33sec



- ① orchestrates for resources
- ② progress of each of the stage
- ③ In case of failures it will retry the stage that failed
- ④ orchestrates entire life cycle

Visualization of Production System



Use Case

Flipkart Handling WannaCry Ransomware

apply patch



12th May, 2017

June Beginning

Microsoft gave a
patch

Restart the
machine

① When should I apply patch and perform restart?

① Least Traffic apply patch

Sayon Flipkart → COO

get one time least traffic

nginx → server logs how much

traffic was there at

any hour of our day

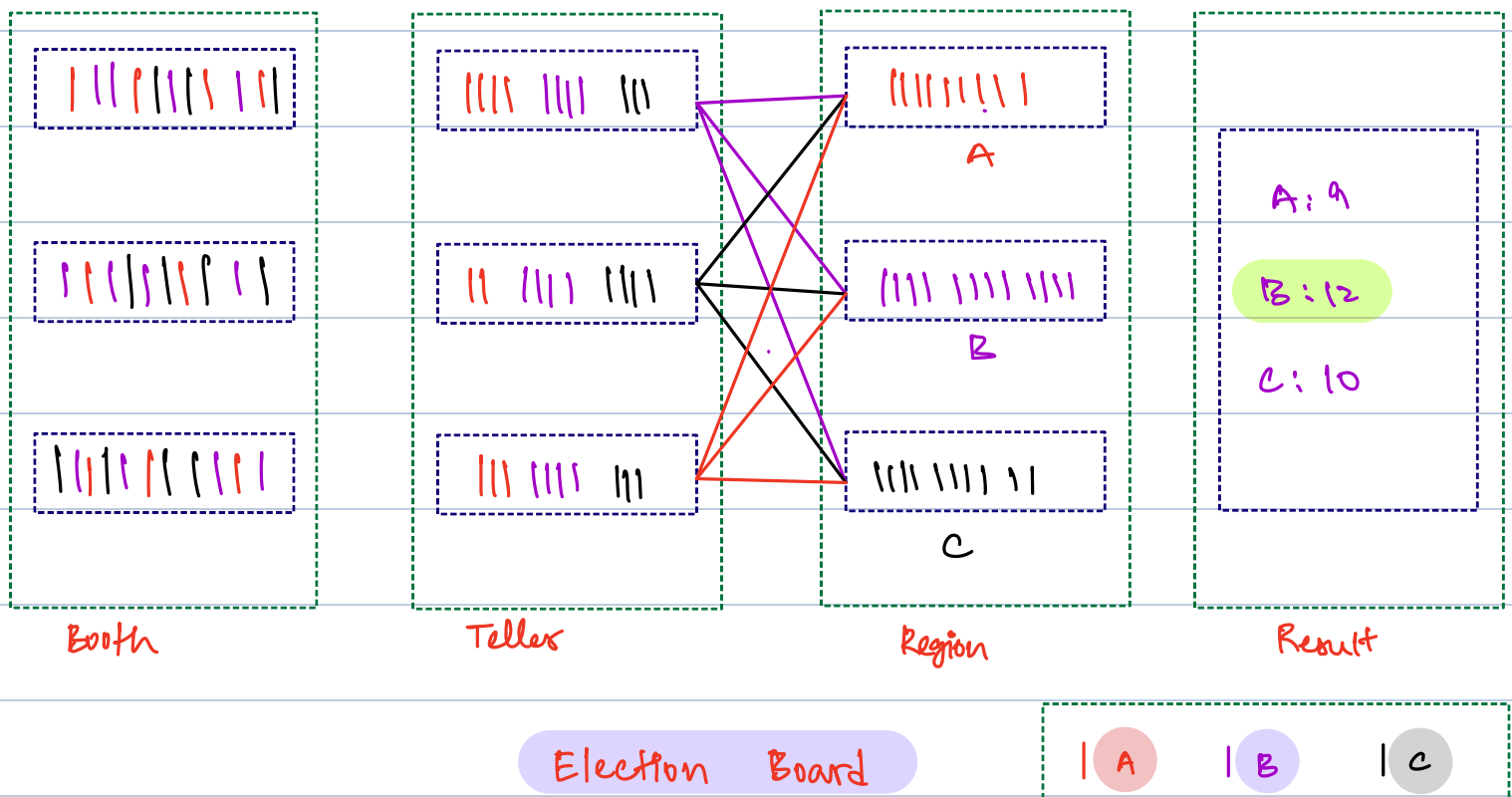
Map Reduce

What is Map Reduce ?

programming construct for handling large datasets

- 2 parts
- ① Map Input $\langle \text{key}, \text{value} \rangle$ output $\langle \text{key}, \text{value} \rangle$
 - ② Reduce Input $\langle \text{key}, \text{value} \rangle$, output final ans

How is Map Reduce Implemented ?



Distributed data $\rightarrow$ mapper $\langle \frac{\text{key}}{A} \quad \frac{\text{value}}{4} \rangle$
partition

Phases in Map Reduce



Map phase

- Reads assigned input split from HDFS
- Parses input into records as key-value pairs
- Applies map function to each record
- Informs master node of its completion



Partition phase

- Each mapper must determine which reducer will receive each of the outputs
- For any key, the destination partition is the same
- Number of partitions = Number of reducers



Shuffle phase

- Fetches input data from all map tasks for the portion corresponding to the reduce tasks bucket



Sort phase

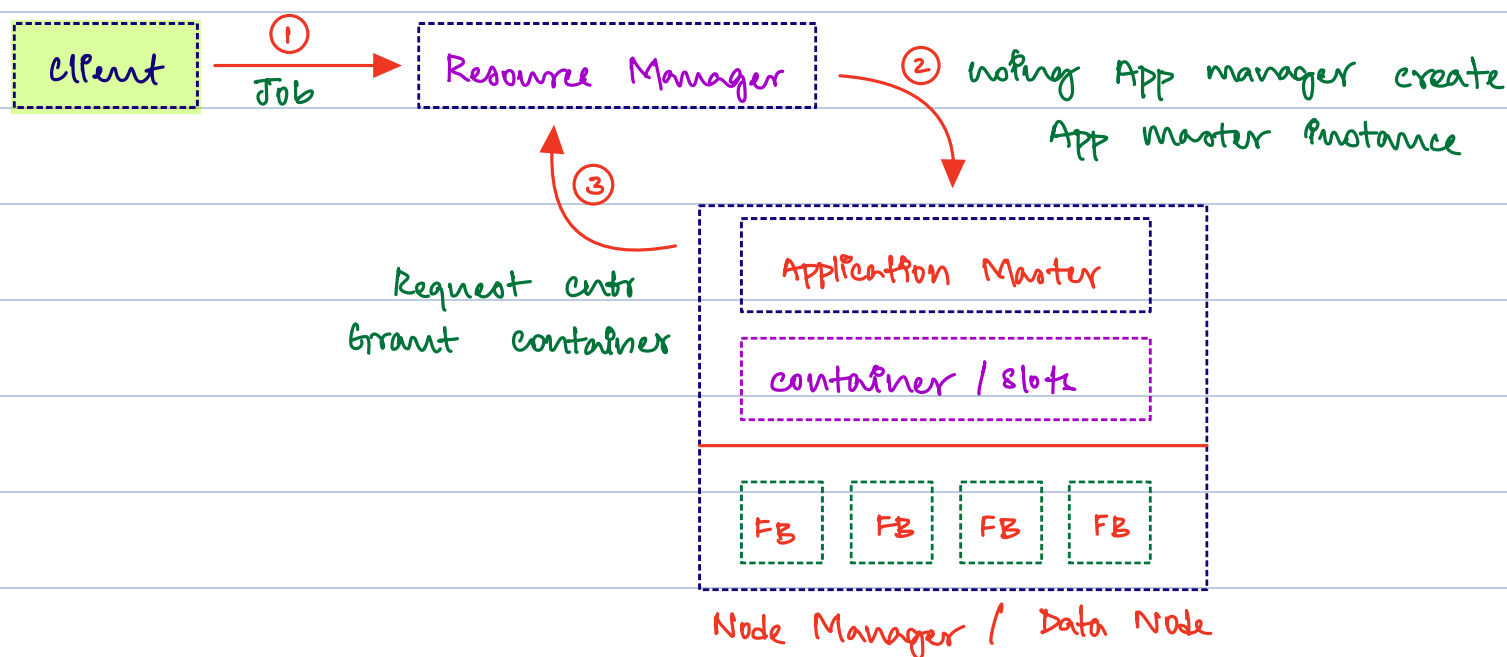
- Merge sorts all map outputs into a single run



Reduce phase

- Applies user-defined reduce function to the merged run

How is map reduce executed



④ Launches Mapper logic on Node Manager

mapper / partitions

⑤ Runs shuffle / sort sequentially on Node Manager

⑥ Runs Reducer

Friday's class

- ① Map Reduce Internals
- ② The code of mapper and reducer
- ③ How of submitting the job